

Part 1 Topics Guide: Regression Foundations

What you should be able to explain after Units 01–07. These are stylized facts, not derivations. If you can say each bullet in your own words and give an example, you are ready for Part 2.

Intuitive formulas worth remembering	
Slope = $\text{cov}(x, y) / \text{var}(x)$ (“how much Y moves per unit of X”) Correlation = $\text{cov}(x, y) / (\text{sd}(x) \cdot \text{sd}(y))$ (“how tightly”) R² = explained variance / total variance of Y SE of the mean $\approx \text{sd}(x) / \sqrt{n}$ SE of the slope $\approx \text{residual} / (n \cdot \text{var}(x))$	Confidence interval = estimate $\pm 2 \times \text{SE}$ ($t \approx 1.96$ for 95%) Naïve slope = true slope + bias (direct + indirect effect) Omitted variable bias = $\alpha_1 \times \beta_2$ = (effect of X on the omitted Z) $\times$ (effect of Z on Y) Diff-in-diff b_3 = (treated after – before) – (comparison after – before)

01 • Nuts and Bolts of Regression

- **Everything is built from the deviation** (a point’s distance from the mean). Variance, covariance, correlation, and the slope are the same operation with different denominators.
- **Variance is a total; standard deviation is an average** — the typical distance from a point to the mean, in the original units.
- **Covariance works by signs**: points in the (+,+) and (–,–) quadrants push it up, the others push it down. One outlier can move it a lot.
- **Slope $\neq$ strength**. The slope keeps units and answers *how much*; correlation drops units and answers *how tightly*. A steep line can be a weak relationship.
- **The regression line is a conditional mean** — the mean of Y once it is allowed to depend on X. It always passes through $(\bar{x}, \bar{y})$.
- **A residual is what the model missed**: observed Y minus predicted Y.
- **Total variance = explained + unexplained** (TSS = regression SS + error SS). R^2 is the explained share. Watch the abbreviations: textbooks swap RSS and ESS.

02 • Variance, Standard Errors, and Confidence Intervals

- **Two distributions**. The *sample* distribution is your data (spread = standard deviation). The *sampling* distribution is the estimates you would get from many samples (spread = standard error). In the second picture, one dot is a whole study.
- **The standard error is the “average error”** — how far a typical estimate lands from the truth. Your regression line is one draw, not the true line.
- **Three levers shrink the SE of a slope**: less residual noise, more cases, more variation in X. **Noise in Y hurts; spread in X helps**.
- **Sample size has diminishing returns**: quadruple n to halve the SE.
- **The Central Limit Theorem** makes the sampling distribution normal, which is what lets us turn an SE into an interval.
- **The 95% belongs to the procedure**. The true slope is fixed; the interval is what moves. Your one interval either contains the truth or it does not.
- **Significance is downstream of fit**: unexplained variance $\rightarrow$ SE $\rightarrow$ confidence interval $\rightarrow$ p-value.

- **Significance only certifies the sign**. A CI that excludes zero tells you the direction of the effect, not its size or whether the program is worth the cost.
- **A program is a bet**: the coefficient is the expected payoff, the CI is the risk. The “insignificant” program can be the better bet.
- **The p-value is a threshold, not a property of the finding**. The same estimate can be significant at 95% and not at 99%.
- **Good estimates are unbiased (accurate) and efficient (precise)**. These are separate goals.

03 • Control Variables

- **Picture the Venn (Ballentine) diagram**: the overlap of X and Y is the explained part; the rest of Y is the residual. Slope $\approx \text{overlap} / \text{var}(X)$; $\text{SE} \approx \text{residual} / \text{var}(X)$.
- **A control works on one of two regions**: the residual or the overlap. Which one decides everything.
- **Controls uncorrelated with the treatment** shrink the residual: **slope stays the same, SE gets smaller**. They buy precision almost for free.
- **Controls correlated with the treatment** take a share of the overlap: **slope changes, SE usually grows**. They buy accuracy at the cost of precision.
- **What matters is the control’s correlation with X**, not with Y. The ideal control predicts Y and is unrelated to the treatment.
- **As a control approaches perfect correlation with X, the SE blows up** — no independent variation is left.
- **Widening the range of the treatment** raises $\text{var}(X)$ and shrinks the SE, but only at the design stage.

04 • Omitted Variable Bias

- **Greek letters are the truth, Latin letters are our guess**. The full model gives β_1 ; the naïve model gives b_1 .
- **Bias = $\alpha_1 \times \beta_2$** . The omitted variable must be related to *both* the treatment and the outcome. If either link is zero, there is no bias.
- **The naïve slope is the direct effect plus the indirect effect** through the omitted variable. Regression cannot separate routes it cannot see.
- **You can often sign the bias** from theory (sign of $\alpha_1 \times$ sign of β_2) without measuring the variable, and so tell whether your estimate is too big or too small.
- **Bias does not shrink with sample size**. It is a missing *column* problem, not a missing *row* problem.

- **The accuracy/precision 2 × 2:** the worst outcome is **biased and precise**, because tight intervals hide the error. Adding noise-reducing controls to a mis-specified model makes it look better without making it better.
- **No diagnostic detects a missing competing hypothesis.** That takes theory.
- **Randomization sets $\alpha_1 = 0$** for every confounder, measured or not. It buys accuracy, not precision, and turns every control into the harmless, uncorrelated kind. Attrition and non-compliance can bring bias back.

05 • Dummy Variables and Interactions

- **A factor with k levels gets k – 1 dummies.** The omitted level is the reference group, and every coefficient is a comparison to it.
- **Changing the reference group changes every coefficient but no fitted line.** R^2 does not move. Choose the reference that matches your research question.
- **Dummies shift intercepts; interactions shift slopes.** A group's slope is the reference slope *plus* its interaction term.
- **Every interaction asks the same question:** is the combination simply additive, or is there something extra? The interaction coefficient is the extra.
- **Diff-in-diff is a dummy × dummy interaction.** b_0 = comparison group before, b_1 = treated group's starting gap, b_2 = common trend, b_3 = **program effect**. The counterfactual is $b_0 + b_1 + b_2$.
- **DiD tolerates level differences between groups but not trend differences** — the comparison group's trend *is* the counterfactual.
- **The design matrix decides which hypotheses get tested**, not how well the model fits. You can always recover every group mean from the coefficients.
- **Pooled comparisons can hide real effects** (Teach for America): within-group comparisons answer the causal question.

06 • Specification and Measurement Error

- **Always plot the data.** Anscombe's quartet: identical summary statistics, four different relationships.
- **For every threat, ask two questions:** what does it do to the slope, and what does it do to the SE?
- **Adding a constant is free** (only the intercept moves). **Adding random noise is not**, because it changes $\text{var}(x)$.
- **Measurement error in Y** lands in the residual: slope unbiased, SE larger, risk of a false negative.
- **Measurement error in X causes attenuation:** the slope is pulled **toward zero** and the SE shrinks — a too-small effect reported with confidence.
- **Outliers in the middle of X inflate the SE; outliers at the extremes of X bend the slope.** Leverage comes from distance along X. Cook's distance flags influence, not error.
- **Logs fix skew; quadratics fix curvature.** Both change what the coefficient means, so restate the interpretation.

07 • Threats to Inference and Statistical Power

- **Seven sins:** omitted variables, multicollinearity, measurement error, misspecification, group heterogeneity, selection, and

simultaneity. Most reduce to one mechanism: **something correlated with X is outside the model**.

- **Multicollinearity is the price of fixing OVB.** It inflates the SEs of both correlated variables; it does not bias them.
- **Pooled and within-group slopes can differ, even in sign** (Simpson's paradox). Fixed effects remove all stable group differences without measuring them. A "group" can be one person measured over time.
- **Selection bias:** ask what drove the choice to participate. That is your omitted variable.
- **Power ≠ confidence.** Confidence is measured against the true slope; power is the chance the CI excludes zero.
- **Power is always power to detect a specific effect** under a specific design. Big effects are easy to find; small ones need large samples. The levers for power are the levers for precision.
- **Type I errors (false positives) usually come from bias**, a modelling failure. **Type II errors (false negatives) usually come from large SEs**, a design failure that cannot be fixed after the data are collected.

Cheat sheet: what each problem does

Situation	Slope	SE
Add control uncorrelated with X	unchanged	smaller
Add control correlated with X	changes (less biased)	usually larger
Omit a variable related to X and Y	biased ($\alpha_1\beta_2$)	often smaller (false confidence)
Multicollinearity	unbiased	larger
Measurement error in Y	unbiased	larger
Measurement error in X	toward zero	smaller
Outlier in the middle of X	unbiased	larger
Outlier at an extreme of X	too steep or too flat	larger
Randomized treatment	unbiased	unchanged
Larger n / wider range of X	unchanged	smaller

The two questions to ask of every estimate: Is it accurate (bias)? Is it precise (standard error)?